Natural Language Processing Project: Twitter Sentiment Classification

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Crisis Detection:

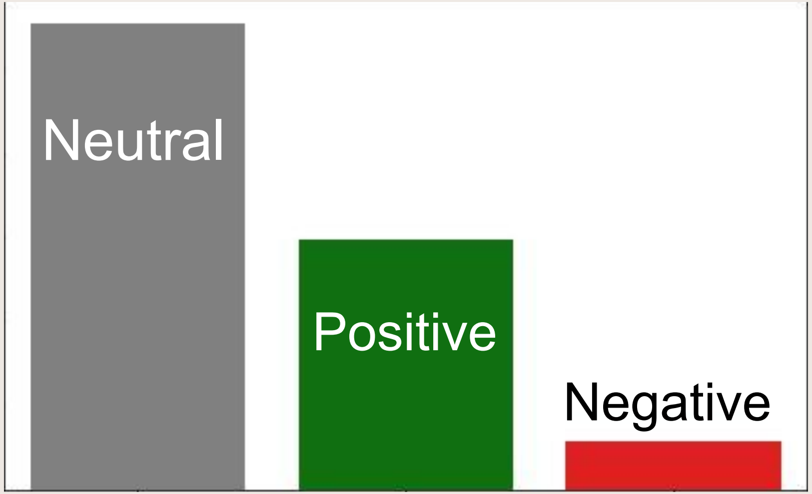
Risk: One negative tweet can start a PR crisis

Problem: Manually scanning thousands of mentions is impossible

Goal: Build a reliable, automated **First-Pass Filter**

Twitter Data

- 9,000+ tweets from 2013 (CrowdFlower Open Source)
- Sentiment labels
- Imbalanced (real-world)



Method: Finding the ‘Lit’ Matches

Cleaned text data

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Created Stopword list

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Ran Dummy Model

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Tested 20+ Model Combinations

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Selected Support Vector Classifier Model

Fine Tuning Parameters:

Negative F1-Score: Compromise of Recall & Precision

Dummy Model: Negative F-1 Score of **0%**

First SVC Model: Negative F-1 Score of **31.8%**

After Tuning: Negative F1-Score increased to **38%** (a 6.2% increase).

Dial 1: Class Imbalance

Dial 2: Regularization/ Stabilize

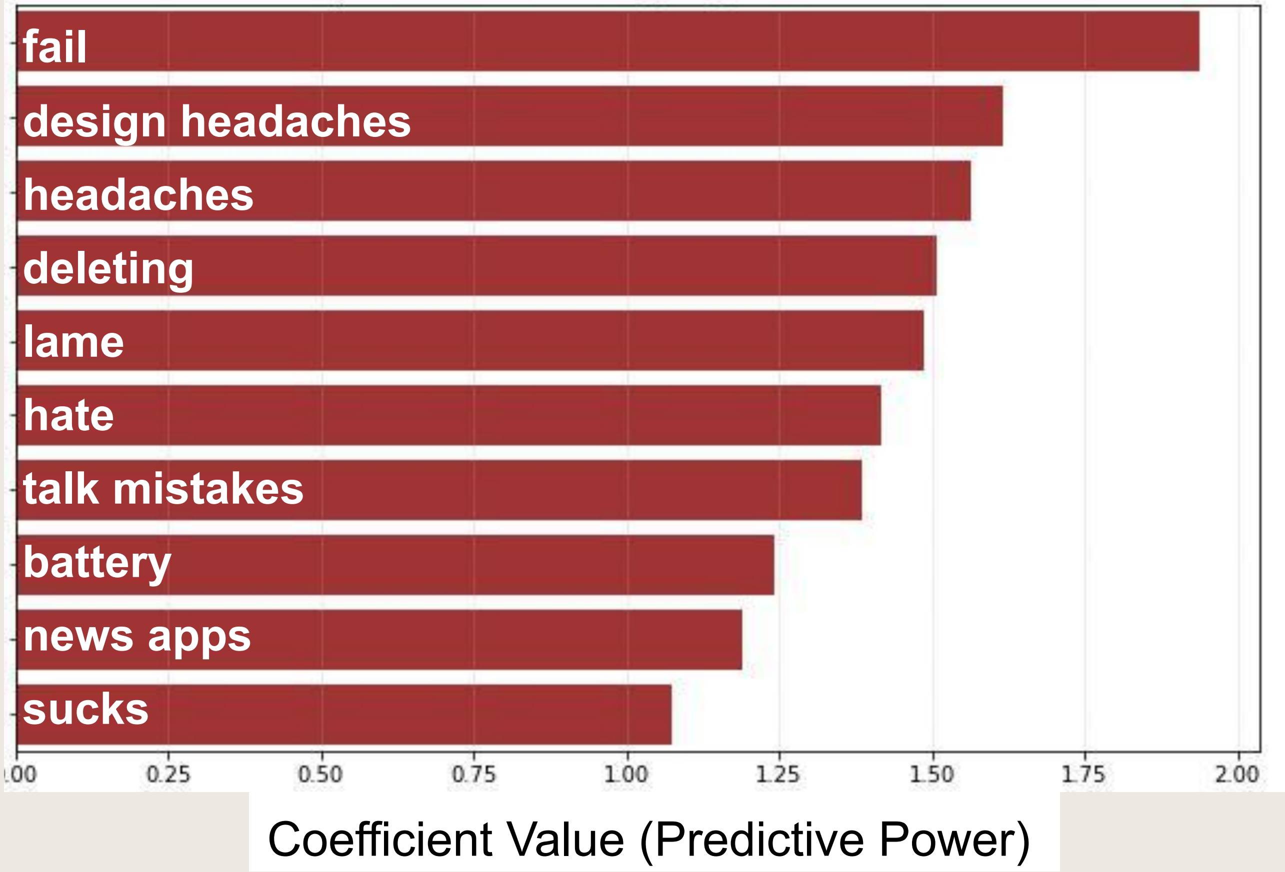
Final Model Results: Ups & Downs

UP: Trustworthiness: Negative Precision hit 50%, cutting false alarms in half.

UP: Improved Performance: Negative F1-Score increased to 38%

DOWN: Remaining Risk: Negative Recall is 30%, meaning most crises are still missed.

Top 10 Features Driving Negative Sentiment



Next Steps:

1: Increase Recall: Explore advanced techniques (e.g., synthetic data, oversample minority) to catch more than 30% of true crises.

2: Acquire More Data: Need more data that is recent, and labeled

3: Immediate Deployment: Use the current model for testing

New Opportunity: Use model to identify positive traction for proactive marketing

Thank you. Questions?